Thi Online KTMT&HN nhóm lẻ

Started on	Friday, 31 May 2019, 1:15 PM
State	Finished
Completed on	Friday, 31 May 2019, 2:25 PM
Time taken	1 hour 10 mins

Question 1
Complete
Marked out of 1.00

For better speed, in CPU design, engineers make use of the following techniques:

Select one or more:

- [ ] Speculative execution
- [x] Branch prediction
- [ ] Pipelining
- [x] Faster CPU internal bus

Question 2
Complete
Marked out of 1.20

What is the correct sequence of instruction cycle?

Step 3 — Fetch operand
Step 6 — Store result
Step 4 — Calculate operand address
Step 1 — Fetch opcode
Step 5 — Execution
Step 2 — Decode

Question 3
Complete
Marked out of 1.00

Part of computer memory is shown in figure

Address	1D48	1D49	1D4A	1D4B	1D4C	1D4D	1D4E	1D4F
Value	03	7F	F5	2D	5A	12	7B	C0

What is the value of AX register after instruction **MOV AX, [1D4B]** executed

Answer: 5A2D

studeersnel
messages.pdf_cover_qr_code_label
messages.downloaded_by

Question 4
Complete
Marked out of 1.00

the memory stack area of aprogramshowninfigure

Address	1D50	1D51	1D52	1D53
Value	AF	90	71	DA

The value of SP register is 1D50. What isthe value of SP follows the execution of **PUSH SI**

Answer: 1D4F

Question 5
Complete
Marked out of 0.50

The instruction that loads the AH register with the lower byte of the flag register is

Select one:

- LAHF
- PUSHF
- SAHF
- AH

Question 6
Complete
Marked out of 1.00

Select correct definition of seek time, rotational delay, access time, transfer time for hard drives with moveable-head system:

rotational delay: transfer time

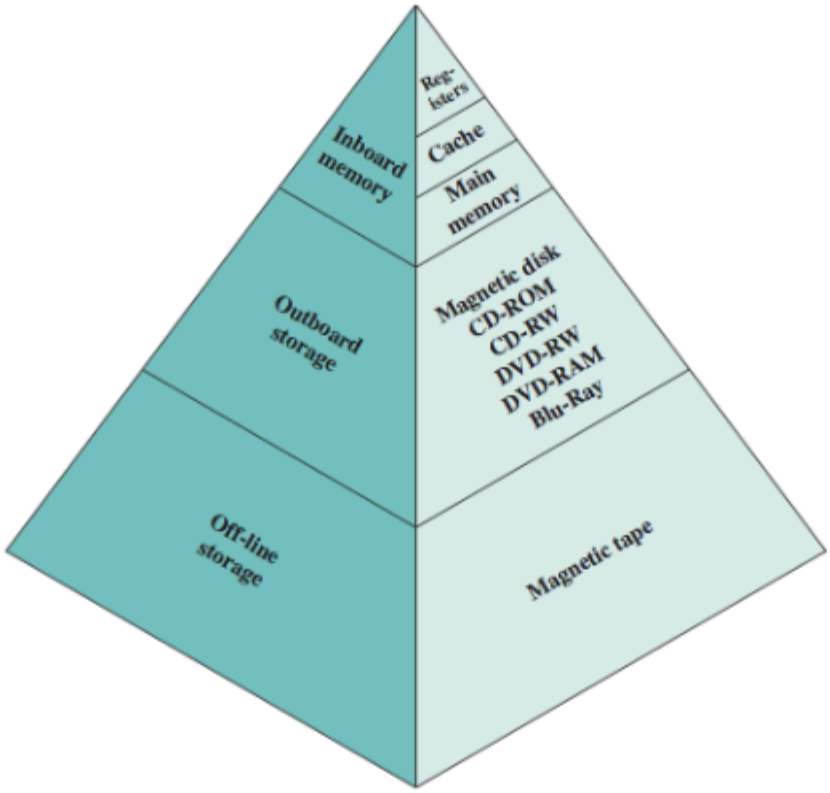
access time: seek time + rotational delay

seek time: time for the sector in the request track to reach the head

studeersnel
messages.pdf_cover_qr_code_label
messages.downloaded_by

Question 7
Complete
Marked out of 1.00

For memory hierarchy below, whichrelationshipholdwhenmovingdownward



Select one or more:

- [x] Decreasing frequency of access by the processor
- [x] Increasing capacity
- [x] Decreasing cost per bit
- [x] Increasing access time
- [ ] the processor accesses more often

Question 8
Complete
Marked out of 0.50

Sign-extend number 1011 0101 (8-bit binary) to 16-bit

Answer: B5

Question 9
Complete
Marked out of 1.00

In computer organization, the CPU transfer rate is much higher than that of memory. It is easy to match performance of these components by:

Select one:

- ( ) increase the bus speed
- ( ) increase I/O speed
- ( ) Introducing cache memory
- (•) producing faster memory module

studeersnel
messages.pdf_cover_qr_code_label
messages.downloaded_by

Question 10
Complete
Marked out of 2.00

Choose correct RAID volume definitionsforarequest2Tstorage.

RAID 0 - Striped volume	2 x 1T HDDs are needed, enhance data transfer, no fault tolerance, data lost when one HDD fails
RAID 1 - Mirror volume	2 x 1T HDDs are needed, no data lost when the primary storage fails
RAID5 Volume	2T HDD + more HDDs to extend storage, no fault tolerance, data lost when one HDD fails
Spanned Volume	At least 3 x 2T HDDs, fault-tolerance, no data lost, no down-time

Question 11
Complete
Marked out of 0.50

Which are the correct inputs for XLAT instruction

Select one or more:

- [ ] look-up index must be loaded into DL
- [x] DS:[BX] pointed to look-up table
- [ ] DS:[SI] pointed to look-up table
- [ ] look-up index must be loaded into AL

Question 12
Complete
Marked out of 1.00

The principle of cache memory relies on key features: locality of reference which involves spatial and temporal locality. Match the definition to keywords on the left

	tendency to use large cache and prefetch mechanism
Spatial locality	the tendency for a processor to access memory locations that have been used recently
Temporal locality	the tendency of execution to involve a number of memory locations that are clustered

Question 13
Complete
Marked out of 1.00

The following sequence of instructions are executed. What is the correct value of AX, CX, DX at watch point?

```
MOV AX,0020
MOV CX,0010
MUL CL
```

watch point:

AX =	0200
DX	00FF
CX =	0010

studeersnel
messages.pdf_cover_qr_code_label
messages.downloaded_by

Question 14
Complete
Marked out of 1.00

Which statements are correct forHDDs?

Select one or more:

- [x] a. Head, Track, Sector are key parameters for access data on hard disk
- [x] b. Bits are stored on tracks
- [ ] c. Bits are store randomly on disk surfaces
- [ ] d. Head, Track, Cylinder are key parameters for access data on hard disk

Question 15
Complete
Marked out of 0.50

In multiplication instruction, when the source operand is 8 bit, _____ will be multiplied with source.

Select one:

- ( ) Whatever general purpose register
- (•) AL
- ( ) AX
- ( ) BX

Question 16
Complete
Marked out of 0.50

Which are the correct actions for LODSW string operation if DF is reset (=0)

Select one or more:

- [ ] Load 16-bit value at memory location pointed by ES:[DI] into AX
- [x] Load 16-bit value at memory location pointed by DS:[SI] into AX
- [ ] increase SI by 2
- [x] decrease DI by 2

Question 17
Complete
Marked out of 0.50

To encrypt a byte value, use _____ instruction.

Select one:

- ( ) OR
- (•) AND
- ( ) NOT
- ( ) XOR

Question 18
Complete
Marked out of 1.00

Which ones are not correct for static RAM?

Select one or more:

- [x] Cheaper than dynamic RAM because simpler chip controller
- [ ] faster than dynamic RAM because they are made from capacitor
- [x] Cost per bit is lower than dynamic RAM
- [ ] Cost per bit is higher than dynamic RAM

studeersnel
messages.pdf_cover_qr_code_label
messages.downloaded_by

Question 19
Complete
Marked out of 0.50

the instruction, CMP to comparesourceanddestinationoperandsby_____

Select one:

- comparing
- adding
- subtracting
- dividing

Question 20
Complete
Marked out of 1.00

Select the correct sequence of instructions to compute -1024/128 (all values are in hex).

Step 1: MOV CX,80

Step 2: MOV CL,80

Step 3: IDIV CX

Step 4: IDIV CL

Question 21
Complete
Marked out of 1.00

Which of the following instructions are not legal addressing?

Select one or more:

- MOV CX, [SI]
- MOV AX, [SP+1]
- MOV AX, [BX+SP]
- MOV AX, [DI]

Question 22
Complete
Marked out of 1.00

Consider the following assembly instruction sequence

```
        XOR BX, BX
        CMP DL, 5
        JLE   a_label
        CMP    DL,17h
        JGE    a_label
        MOV BX, 10h
a_label:
        INC BX
watch point:
...
```

Choose correct value of BX register at watch point for different value of DL?

DL=10 11h

DL=0FFh 01h

DL=0Ah 01h

DL=17h 18h

studeersnel
messages.pdf_cover_qr_code_label
messages.downloaded_by

Question 23
Complete
Marked out of 1.00

Select correct level for contemporarycomputermultilevelmachine

Level 1 — Microarchitecture level
Level 2 — Instruction set level
Level 0 — Digital logic level
Level 5 — High level programming language
Level 3 — Operating system level
Level 6 — Applications
Layer 4 — Assembly Language level

Question 24
Complete
Marked out of 1.50

Which are correct about 32 bit index registers of IA-32 processors:

Select one or more:

- [x] EDI: 32 bit pointer to destination memory in data movement instructions
- [x] ESI: 32 bit pointer to source memory in data movement instructions
- [ ] ESH,EDH: 16 bit pointers to higher memory above 1M
- [x] DI: 16 bit pointer to destination memory in data movement instructions
- [x] SI: 16 bit pointer to source memory in data movement instructions

Question 25
Complete
Marked out of 1.00

Structural components of computer include:

Select one or more:

- [ ] DMA
- [x] Memory
- [ ] Interrupt
- [x] System interconnection
- [x] Central processing unit
- [x] I/O

studeersnel
messages.pdf_cover_qr_code_label
messages.downloaded_by

Question 26
Complete
Marked out of 1.00

A benchmark program is run on a 40 MHz processor. The executed program consists of 100,000 instruction executions, with the following instruction mix and clock cycle count:

Instruction Type	Instruction Count	Cycles per Instruction
Integer arithmetic	45,000	1
Data transfer	32,000	2
Floating point	15,000	2
Control transfer	8000	2

Calculate the execution time for this program.

Given:

$$T = I_c \times CPI \times \tau$$

$$CPI = \frac{\sum_{i=1}^{n}(CPI_i \times I_i)}{I_c}$$

Answer: 16.3

Question 27
Complete
Marked out of 1.00

What is the correct value of SI, AL (in hex) at watch point:

```
01:             MOV SI, 300h
02:             MOV AL, 10h
03:             MOV CX, 7
04: Loop_label:
05:             MOV [SI], AL
06:             ADD AL,10h
07:             INC SI
08:             LOOP Loop_label
```

watch point:

SI 307h

AL = 80h

Question 28
Complete
Marked out of 1.00

Convert 0.1015625 to IEEE 32-bit floating point format (1 sign+ 8 exponent + 23 mantissa)

Answer: F7F49

Question 29
Complete
Marked out of 1.00

Which is correct about dual-layer DVD?

Select one:

- the same as double-sided DVD
- DVD drives has double laser head for reading from or writing to this disk
- contains layers on both sides of the disk for writing data to
- contains two layers on a single side for writing data to

studeersnel
messages.pdf_cover_qr_code_label
messages.downloaded_by

Question 30
Complete
Marked out of 0.50

In multiplication instruction, theupperhalfoftheresultisnonzeroimplieswhich state of Carry flag and Overflow flag?

Select one or more:

- [ ] CF=0
- [ ] CF=1
- [x] OF=0
- [x] OF=1

Question 31
Complete
Marked out of 1.00

Given 8-bit floating-point binary format:
1 (sign) + 3 (exponent) + 4 (mantissa)
Convert the 8-bit floating point number 68 (in hex) to decimal.

Answer: 104

Question 32
Complete
Marked out of 0.50

Which of the following instructions are not valid?

Select one or more:

- [ ] MOV AX, SI
- [ ] MOV AX, [BP+2]
- [x] MOV DS, B800h
- [x] MOV SP, SS:[SI+2]

Question 33
Complete
Marked out of 1.00

Consider a 16-bit microprocessor, with a 16-bit external data bus, driven by an 10-MHz input clock. Assume that this microprocessor has a bus cycle whose minimum duration equals four input clock cycles. What is the maximum data transfer rate across the bus that this microprocessor can sustain?

Select one:

- ( ) 1 MB/s
- (•) 5 MB/s
- ( ) 4 MB/s
- ( ) 10 MB/s

Question 34
Complete
Marked out of 1.00

The following sequence of instructions are executed. What is the correct value of flag bits at watch point?

```
MOV AL, 0F
ADD AL, F1
```

watch point:

Zero flag (OF) = set ▾

Carry flag (CF) = set ▾

studeersnel
messages.pdf_cover_qr_code_label
messages.downloaded_by

Question 35
Not answered
Marked out of 1.00

Convert the 32-bit floating pointnumberC4361000(inhex)todecimal.

Answer:

Question 36
Complete
Marked out of 1.00

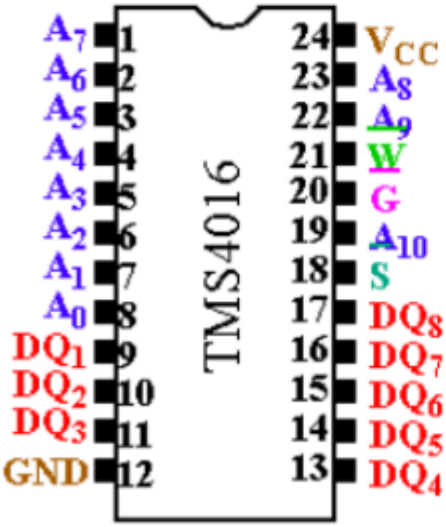
Choose correct features for SRAM and DRAM

SRAM: Faster access time, cost more per bit, smaller size

DRAM: Slower access time, cheaper cost per bit, can manufacture with larger size

Question 37
Complete
Marked out of 1.00

Choose the correct structure of memory chip as shown below



Note:
DQ: Data pinout

Select one:

- SRAM 2Kx8-bit
- DRAM 2Kx8-bit
- SRAM 1Kx16-bit
- DRAM 1Kx16-bit

Question 38
Not answered
Marked out of 0.50

Write mask byte (in hex) to set bit 6th, 4th of a byte value with OR instruction (LSB is the 1st bit).

Answer:

studeersnel
messages.pdf_cover_qr_code_label
messages.downloaded_by

Question 39
Complete
Marked out of 0.50

8088 is 16 bit processor, the maximumaddressablememoryis:

Select one:

- 640K
- 640M
- 1024K
- 64M

Question 40
Complete
Marked out of 1.00

A memory chip has 12 address pins, determine the maximum memory words of this chip?

Select one:

- 2048K
- 4096
- 2048
- 4000

Question 41
Complete
Marked out of 1.50

A benchmark program is run on a 40 MHz processor. The executed program consists of 100,000 instruction executions, with the following instruction mix and clock cycle count:

Instruction Type	Instruction Count	Cycles per Instruction
Integer arithmetic	45,000	1
Data transfer	32,000	2
Floating point	15,000	2
Control transfer	8000	2

Calculate MIPS rate for this program

Given:

$$\text{MIPS rate} = \frac{I_c}{T \times 10^6} = \frac{f}{CPI \times 10^6}$$

$$CPI = \frac{\sum_{i=1}^{n}(CPI_i \times I_i)}{I_c}$$

Answer: 16.3

studeersnel
messages.pdf_cover_qr_code_label
messages.downloaded_by

Question 42
Complete
Marked out of 1.00

To evaluate processor performance,thefollowingindicatorsandformulasareused:

Cycles per instruction $CPI = \frac{\sum_{i=1}^{n}(CPI_i \times I_i)}{I_c}$

Time to execute a program $T = I_c \times CPI \times \tau$

Or $T = I_c \times [p + (m \times k)] \times \tau$

In which:
p: the number of processor cycles needed to decode and execute the instruction
m: the number of memory references needed
k: the ratio between memory cycle time and processor cycle time
τ: cycle time = *1/f*

Which of the following system attributes affects cycle time t

Select one or more:

- [ ] Compiler technology
- [x] Cache and memory hierarchy
- [ ] Instruction set architecture
- [x] Processor implementation

Question 43
Complete
Marked out of 1.50

A system programmer needs to compute 449/2+358/4 (decimal). Instruct him to code in debug (number must be in hex) with the least number of instruction counts.

Step 1: MOV BX,AX
Step 2: MOV AX,01BC
Step 3: ADD AX,BX
Step 4: ADD AX,166
Step 5: SHL AX,CL
Step 6: SHR AX,CL

Question 44
Complete
Marked out of 0.50

Write mask byte (in hex) to clear bit 2nd, 3rd, 5th of a byte value with AND instruction (LSB is 1st bit).

Answer: 10101110

studeersnel
messages.pdf_cover_qr_code_label
messages.downloaded_by

Question 45
Complete
Marked out of 1.00

Consider a magnetic disk drivewith8surfaces,512trackspersurface,and64 sectors per track. Sector size is 1 kB. What is the disk capacity

Answer: 4096 KB

◄ Announcements

Return to: General

studeersnel
messages.pdf_cover_qr_code_label
messages.downloaded_by